# Manuscript 7613110 – Response to reviewers (second revision)

Dear Dr. Mori,

Thank you for the opportunity to submit a minor revision of our manuscript, "Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses," to *Ecology Letters*. We are grateful to the reviewers for their positive assessment of the previous revision and for their remaining suggestions. Our main changes in this revision are:

1. We revised the final paragraph of the Discussion to place our results in the context of Afkhami et al. (2014), noting that endophytes can alleviate drought stress and potentially influence host distributions, but that in our study endophyte-associated hosts showed stronger declines with increasing aridity, as requested by Reviewer 1.

2. We rewrote the Discussion paragraph on symbiont retention to acknowledge that endophyte infections are not always maintained throughout the host life cycle and that environmental conditions can influence their persistence (Afkhami and Rudgers, 2008; David et al., 2019), and we discuss how such loss could affect the interpretation of our results (Reviewer 2).

3. We spelled out genus names at the start of sentences and added the spacing between plants within plots to the Methods (Reviewers 1 & 2).

We reproduce the reviewers' comments below with our point-by-point responses. All changes are denoted in the manuscript with Mahogany font.

On behalf of my coauthors and myself,
Jacob Moutouama

# Response to Reviewer 1

**Comment 1:** *"Aside from one comment (which is merely a suggestion), I feel that all of my comments have been addressed in this version. I have only a couple of minor points, one of style, the other of substance."*

**Response:** We thank Reviewer 1 for the careful second reading of our manuscript and are glad that the previous revision addressed their comments. We respond to the two remaining points below.

**Comment 2:** *"Style: I noted in some of the new (red-highlighted) text instances in which sentences started with abbreviated genera (A. for Agrostis, P. for Poa). In my understanding of typical standards, one shouldn't start a sentence with an punctuated abbreviation, and therefore genus names should be spelled out in full if they are the first word of a sentence."*

**Response:** We thank the reviewer for noting this. We have checked the entire manuscript and now spell out genus names in full whenever they begin a sentence (lines 123–127 and 402).

**Comment 3:** *"Substance: One of my comments (comment 34) asked for speculation about published instances of host-range expansion attributable to Epichloë symbiosis. The response document addresses this acceptably, in my opinion, but I don't see any of that in the text. The response mentions line 434, but that location or more generally that paragraph doesn't explicitly address it. To clarify, I am thinking for example of the finding in: Afkhami ME, McIntyre PJ, Strauss SY (2014) Mutualist-mediated effects on species' range limits across large geographic scales. Ecology Letters 17: 1265–1273. doi 10.1111/ele.12332."*

**Response:** We thank the reviewer for pointing out this omission and apologize that the speculation described in our previous response did not appear in the manuscript. We have now revised the final paragraph of the Discussion to place our results in the context of Afkhami et al. (2014). We note that, although endophytes can alleviate drought stress and potentially influence host distributions, endophyte-associated hosts in our study showed stronger declines in demographic performance with increasing aridity. We also note that the demographic responses we measured may not translate directly into changes in population growth or geographic range, neither of which we estimated, and we identify the consequences of endophytes for host persistence and range expansion under increasing aridity as an open question (line 432).

The revised text reads:

> "Although endophytes can alleviate drought stress and potentially influence host distributions, their effects on host performance are context-dependent (Afkhami et al., 2014). In our study, endophyte-associated hosts showed stronger declines in demographic performance with increasing aridity, suggesting that endophytes do not necessarily buffer hosts against abiotic stress. These demographic responses may not translate directly into changes in population growth or geographic range, neither of which we directly estimated. Whether endophytes promote host persistence or range expansion under increasing aridity, and how climate-driven changes in endophyte prevalence influence these outcomes, remain open questions (Fowler et al., 2025; Li et al., 2026)."

# Response to Reviewer 2

**Comment 4:** *"The authors have addressed the majority of my concerns. Below are my remaining concerns."*

**Response:** We thank Reviewer 2 for their continued engagement with our manuscript. We address the remaining concerns below.

**Comment 5:** *"Lines 177–179: Please include the distance between plants here. I was left wondering if it was possible to measure individual growth or if the plants had begun to grow into each other making it difficult to differentiate between individuals."*

**Response:** We thank the reviewer for this helpful suggestion. Plants were arranged in an evenly spaced $4 \times 4$ grid within each plot, approximately 37.5 cm apart. We have clarified this arrangement in the Methods (line 181).

**Comment 6:** *"Line 389–403: The assertions made in this paragraph around the retention of endophytes are not supported by the literature. As I brought up in my last review, there is a lot of evidence of imperfect transmission between tillers in a single plant and single generation. For example, from the abstract of Afkhami and Rudgers (2008) "Endophytes were lost at all possible stages: within adult plants, from adult tillers to seeds, and from seeds to seedlings." Furthermore, David et al. (2019) found that endophytes were lost more frequently from plants under experimentally reduced precipitation. This paragraph should be written to reflect the results of past studies."*

**Response:** We thank the reviewer for pressing this point, and we agree that our original paragraph overstated the stability of the symbiosis. We have revised the paragraph to acknowledge that endophyte infections are not always maintained throughout the host life cycle and that environmental conditions can influence their persistence (Afkhami and Rudgers, 2008; David et al., 2019). We now discuss endophyte loss under arid conditions as a possible explanation for the weaker demographic benefits observed with increasing aridity. Because we did not track endophyte status throughout the experiment, we cannot determine whether such loss occurred. If plants hosting the symbiont at the start of the experiment subsequently lost it, this could have weakened the measured differences between symbiotic and non-symbiotic plants. We have also identified tracking endophyte status over time as a priority for future research (line 389). The revised text reads:

"Endophyte loss under drought may also contribute to the decline in endophyte benefits with increasing aridity. Endophyte infections are not always maintained throughout the host life cycle, and environmental conditions can influence their persistence (Afkhami and Rudgers, 2008; David et al., 2019). If endophyte loss occurred at the drier sites of our gradient, some plants hosting the symbiont at the start of the experiment may have lost their endophyte, potentially weakening the measured differences between symbiotic and non-symbiotic plants. We did not track endophyte status throughout the experiment and therefore cannot test this possibility directly. The likelihood and consequences of endophyte loss may depend on host–symbiont specialization and functional differences among endophytes (Saona et al., 2010). Future studies should track endophyte status over time and assess host resource status across aridity gradients and host–endophyte systems to determine how these factors contribute to variation in the demographic benefits of symbiosis."

# References


Afkhami, M. E., McIntyre, P. J., and Strauss, S. Y. (2014). Mutualist-mediated effects on species' range limits across large geographic scales. *Ecology letters*, 17(10):1265–1273.

Afkhami, M. E. and Rudgers, J. A. (2008). Symbiosis lost: imperfect vertical transmission of fungal endophytes in grasses. *The American Naturalist*, 172(3):405–416.

David, A. S., Bell-Dereske, L. P., Emery, S. M., McCormick, B. M., Seabloom, E. W., and Rudgers, J. A. (2019). Testing for loss of epichloë and non-epichloid symbionts under altered rainfall regimes. *American journal of botany*, 106(8):1081–1089.

Fowler, J. C., Moutouama, J., and Miller, T. E. (2025). Increasing prevalence of plant-fungal symbiosis across two centuries of environmental change. *Global Change Biology*, 31(11):e70577.

Li, V. W., Fowler, J. C., David, A. S., Strauss, S. Y., Searcy, C. A., and Afkhami, M. E. (2026). Climate variability disrupts microbial mutualism-driven population persistence. *Nature Ecology & Evolution*, pages 1–11.

Saona, N. M., Albrectsen, B. R., Ericson, L., and Bazely, D. R. (2010). Environmental stresses mediate endophyte–grass interactions in a boreal archipelago. *Journal of Ecology*, 98(2):470–479.